

The Processor Families

Processor	Clock Speed	Bus Speed	Form Factor	L1/L2 Cache
Athlon	1.4 GHz	266 MHz	Socket A	128/256 KB
AthlonXP	1900+ (1.6 GHz)	266 MHz	Socket A	128/256 KB
Duron	1.2 GHz	200 MHz	Socket A	128/64 KB
Pentium III	1.2 GHz	133 MHz	Socket 370	32/256 KB
Celeron	1.2 GHz	100 MHz	Socket 370	32/256 KB
Pentium 4	2.0 GHz	400 MHz	Socket 478	20/256 KB

The SpecViewperf test consists of six tests performing various complex OpenGL calculations under a variety of conditions.

The Athlon XP 1800+ completed the entire Photoshop test in 63.1 seconds whereas the Pentium 4, 2 GHz took about 68.3 seconds to complete it. In the ray tracing test, the Athlon XP 1800+ finished rendering the image in 7 minutes 2 seconds. The Pentium 4, 2 GHz took 8 minutes 36 seconds to complete the same.

In the performance segment the results were similar: the sheer power of Athlon's FPU strength was too much for the Pentium III to handle.

Surprisingly, in the value segment we saw the Celeron 1.2 GHz pummelling the Duron 1.1 GHz in the Photoshop test—it

took 76.4 seconds to complete the test as compared to the 84.8 seconds taken by the Duron 1.1 GHz.

MAKING A CHOICE

The processor market is out to spoil you with a wide array of choices. Those days when you could only buy an Intel processor and had just a couple of speed options to pick from are gone. Today you can choose between the older Pentium III and Athlon processors—which still provide dependable performance—and the latest cutting edge chips from Intel and AMD. While most of us feel that variety is the spice of life, these various options can actually become pretty confusing.

First of all, you need to know what

Multimedia Performance



you're going to use the system for. That sometimes can get more difficult as most of us have a tendency to load a lot of applications, but use hardly any of them.

If you use the PC for running Office applications and surfing the Web, then one of the Value processors should suit you quite well. If, on the other hand, you are planning to use some of the heavier professional applications such as Photoshop or Corel Draw, the additional power of one of the performance processors would come in handy.

Those who want to use their PCs for more processor-intensive applications—audio/video encoding, 3D modelling

Processor FAQs

1. How do I check that I got the processor I was promised?
Most vendors won't show you what they put inside your PC, but you can use a popular tool known as WCPUID to find out the exact speed and make of your processor. When your PC starts up, check for the CPU name and speed that the BIOS identifies.

2. What is a retail original boxed packed CPU and what are the advantages of buying one?
When you buy a retail original boxed packed CPU, you get a limited warranty for three years. The warranty does not cover non-boxed products purchased from resellers or otherwise, and it doesn't extend beyond what your system vendor has provided. Both AMD and Intel provide original heatsink-fan combinations with boxed processors.

3. How do I monitor my CPUs health?
All modern motherboards support some form of hardware monitoring which allows you to monitor your CPU's temperature. You can also use Motherboard Monitor (MBM v5.1.02), a popular freeware program. It's easy to configure and if you leave it running in the background, it will tell you the maximum temperatures that your CPU touched while running heavy applications or playing games.

You can also set alarms and ask the program to shut down the PC in case the CPU reaches a particular temperature.

4. How hot is hot?
As a rule, make sure your Processor runs below 55 degrees Centigrade at all times.

If you find that your processor crosses 60 degrees

Centigrade, then you need additional cooling. If there is a difference of more than 10 degrees Centigrade between the idle and full load temperatures (when the CPU is running at 100 per cent utilisation), then you need to check whether there is adequate contact between your heatsink and your processor.

You can improve contact by applying a thin layer of thermal grease between the surface of the processor and the heatsink.

5. Is a 1 GHz processor machine twice as fast as a 500 MHz processor?
No, a 25 per cent system wide performance increase is something that you can only hope for. A 10-20 per cent is something that you will see immediately. This is because the machine's performance is based on a number of factors other than just the processor.

6. Can I overclock?
Overclocking means increasing the rated speeds of the processor by changing the internal multiplier or the external frequency. Most modern processors have their internal multipliers locked leaving you the option of only changing the external frequency. By changing the FSB (Front Side Bus) you are in effect also overclocking all the other PCI devices. Overclock your processor only if you understand processors quite well and are willing to take the risk of the warranty being void.

Among the tested processors, the Pentium 4, 2 GHz could be overclocked to 2.34 GHz and it ran flawlessly even at over 340 MHz above its rated frequency. The new Tualatins, the Celeron 1.2 GHz, and the Pentium III 1.13 GHz are also overclocker-friendly.